Journal of Energy Storage 84 (2024) 111001

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

Research papers

Frequency reconstruction oriented EMD-LSTM-AM based surface temperature prediction for lithium-ion battery

Xiao Qi [a], Chaofeng Hong [a], Tao Ye [a], Lijun Gu [b,*], Weixiong Wu [a,*]

[a] Energy and Electricity Research Center, Jinan University, 519070, Guangdong, China
[b] College of Mechanical and Control Engineering, BaiCheng Normal University, 137000, Jilin, China

ARTICLE INFO

Keywords:
Lithium-ion batteries
Long short-term memory
Frequency reconstruction
Battery temperature prediction

ABSTRACT

With the development of electric vehicles, safety concerns, especially thermal runaways, have garnered widespread attention. Accurate temperature prediction is essential to avoid the occurrence of thermal runaway. Herein, this paper proposes an innovative frequency reconstruction oriented EMD-LSTM-AM based surface temperature prediction method for lithium-ion batteries, which combines empirical mode decomposition (EMD), long short-term memory (LSTM), and attention mechanism (AM). First, this method employs EMD to extract the overall data trends from the raw data and restructures them into three different frequency components. Subsequently, the LSTM-AM network is utilized to learn long-term dependencies within the features and predict future temperatures. Finally, the accuracy and robustness of the proposed prediction method were validated under five distinct environmental temperatures and four complex driving conditions. The experiments indicate that, at 25 °C, the temperature prediction maximum absolute error achieved by the proposed network is 0.31 °C, exhibiting a 0.23 °C improvement compared to LSTM. Furthermore, under various operating conditions and temperatures, the proposed network demonstrates performance improvements across different metrics. These results suggest that the method effectively handles high-frequency non-stationary noise, mitigates prediction deviations caused by changes in battery physical characteristics, and provides a reliable reference for battery thermal management systems.

1. Introduction

1.1. Motivation

As global fossil fuel reserves continue to diminish, electric vehicles (EVs) have garnered extensive usage thanks to their zero-carbon emissions and their potential to replace conventional gasoline-powered automobiles [1,2]. Lithium-ion batteries (LIBs) are extensively utilized as the primary power source for electric vehicles due to their high energy density and rapid charge–discharge capabilities, among other advantages [3,4].

During the actual operation of EVs, the complex interplay of various operational conditions results in notable changes in the internal parameters of LIBs, including aspects such as internal electrochemical reactions, temperature distribution, and internal resistance [5]. These changes further emphasize the nonlinear polarization effect of the battery, resulting in increased internal resistance and elevated heat generation [6]. As heat generation increases, the operating temperature of LIBs is expected to rise significantly. In the event of inadequate heat dissipation and uncontrolled temperature rise, there is a substantial risk of thermal runaway occurring [7]. Consequently, a responsive and accurate temperature monitoring system is one of the most crucial functions of a battery management system.

Additionally, under the real driving conditions of EVs, the internal battery temperature exhibits rapid fluctuations, necessitating timely temperature monitoring. To achieve this, short-term prediction models are required to capture temperature fluctuations. However, short-term predictions demand high-frequency sampling and excessively high sampling rates can introduce interference from high-frequency non-stationary noise. Therefore, developing lithium-ion battery temperature prediction models with higher prediction frequencies and greater accuracy is a pressing challenge.

1.2. Literature review

Currently, there are two primary model categories employed to study battery behavior: physical models [8,9] and data-driven models

* Corresponding author.
E-mail addresses: qixiao.jnu@gmail.com (X. Qi), chaofengh@126.com (C. Hong), strong_tao@163.com (T. Ye), gulijun@bcnu.edu.cn (L. Gu), weixiongw@126.com (W. Wu).

https://doi.org/10.1016/j.est.2024.111001
Received 11 October 2023; Received in revised form 18 January 2024; Accepted 15 February 2024
Available online 20 February 2024
2352-152X/© 2024 Elsevier Ltd. All rights reserved.

Nomenclature

Acronyms

AM	Attention mechanism
BMS	Battery management system
BP	Back propagation
CCCV	Constant-current constant-voltage
EMD	Empirical mode decomposition
EVs	Electric vehicles
HWFET	Highway fuel economy driving schedule
IMFs	Intrinsic mode functions
LA92	Los Angeles 92
LIBs	Lithium-ion batteries
LSTM	Long short-term memory
MAE	Mean absolute error
MAXE	Maximum error
RMSE	Root mean square error
RNNs	Recurrent neural networks
SoC	State of charge
ST	Surface temperature
SVM	Support vector machine
UDDS	Urban dynamometer driving schedule
US06	Supplemental federal test procedure driving schedule

Symbol

σ	Sigmoid function
$\hat{y}_i$	Predicted value
h_{t-1}	Previous time step output
N	Dataset length
$r(t)$	Residual term
v	Attention value
$\tilde{C}_t$	Output of the tanh layer
b_c	Candidate input gate bias vector
b_f	Forget gate bias vector
b_h	AM bias vector
b_i	Input gate bias vector
b_o	Output gate bias vector
C_t	New memory cell state
C_{t-1}	Previous memory cell state
f_t	Output of the forget gate
h_t	Current time step output
i_t	Output of the input gate
$imf_i(t)$	Intrinsic mode function
W_c	Candidate input gate weight matrix
W_f	Forget gate weight matrix
W_h	AM weight matrix
W_i	Input gate weight matrix
W_o	Output gate weight matrix
x_t	Current time step input
y_i	Actual value

[10,11]. Physical models are constructed based on the physical characteristics of batteries, but overly simplified physical models may fail to accurately depict the temperature distribution within batteries, while more complex models pose computational demands and challenges in acquiring internal battery parameters [12]. In comparison, data-driven methods solely rely on externally measured data without the need to understand the internal structure or parameters of the battery. They have advantages in terms of universality and portability, making them highly suitable for online predictions within the context of battery management systems (BMS) [13]. As a result, data-driven models have garnered increasing attention from researchers both domestically and internationally.

Machine learning is an essential approach within the realm of data-driven models [14]. Compared to traditional physical models, machine learning methods have been found to achieve comparable or even better performance in battery state detection [15], battery capacity prediction [16], and state-of-health estimation [17]. For instance, in article [18], a support vector machine (SVM) model was proposed, which can be implemented in a low-cost microcontroller-based BMS system. Furthermore, in article [19], an improved back propagation (BP) neural network algorithm was introduced, utilizing battery current, terminal voltage, and environmental temperature as inputs, offering fast calculations, and significantly reducing battery state of charge (SoC) prediction errors in practical applications. Additionally, in the article [20], a long short-term memory (LSTM) model was presented, which encodes feature behavior under different environmental temperatures within the network while maintaining high prediction accuracy across these varying temperature conditions.

Similarly, machine learning-based approaches also provide an alternative solution for battery temperature estimation. In the article [21], an artificial neural network was constructed to predict battery surface temperatures at different constant charging rates under varying environmental temperatures. In the article [22], a temperature prediction model for solar cells was developed using a particle swarm optimization support vector machine. Nevertheless, these network structures cannot remember previous information, which can lead to prediction biases due to the close relationship between changes in battery surface temperature and the previous time step. To overcome this issue, the article [23] proposed a recurrent neural networks (RNNs) model that does not require the measurement of environmental temperatures to estimate battery surface temperatures. Whereas, simple RNNs cannot capture long-term dependencies due to the vanishing or exploding gradient problem during the backpropagation process [24]. Therefore, LSTM neural networks with gating systems were introduced. In the article [25], a battery temperature prediction model based on the LSTM algorithm was developed, which can predict temperatures at a specific time point and forecast temperature fluctuations over a longer period. The article [26] introduces convolutional neural networks and a PID compensator on the basis of LSTM networks, enhancing the predictive performance of the model for real-time data input. In article [27], the author employs three different models to analyze the impact of network structure and modeling techniques on temperature prediction. The study demonstrates that the Elman neural networks model is a preferable choice for predicting battery temperature. However, the above model has not been validated under complex operating conditions. As discussed in article [28], accurately assessing the variation in battery heat generation rates under complex driving conditions is a significant challenge. This challenge ultimately makes it difficult to accurately capture the surface temperature of automotive batteries in complex operational scenarios.

1.3. Contribution

To mitigate the interference from high-frequency non-stationary noise in short-term predictions and enhance prediction accuracy, this paper proposes a surface temperature prediction method for LIBs based on EMD-LSTM-AM with a focus on frequency reconstruction. The effectiveness of the method has been verified under various temperature conditions and complex operating scenarios. This method provides a new theoretical approach to battery temperature prediction and offers guidance for the design of battery management systems. The main contributions of this study are as follows.

- The proposed LSTM-AM architecture reduces the dimensionality of hidden layer output data, thereby reducing network parameters while preserving temporal information. This improvement enhances computational speed without any sacrifice in prediction accuracy.
- The proposed frequency-domain decomposition and reconstruction method preprocesses the data, alleviating the interference from high-frequency non-stationary noise in short-term predictions. Moreover, it effectively harnesses spectral biases [29] in deep learning, reducing cumulative errors within the model and enhancing overall accuracy.
- The proposed method was trained and validated under various dynamic driving profiles and temperature conditions. Experimental results indicate that the proposed network achieves high-precision predictions under all tested conditions.

The rest of this paper is organized as follows. Section 2 provides a detailed exposition of the proposed temperature prediction algorithm and its underlying principles. Section 3 introduces the data source and outlines the evaluation metrics used in this study. In Section 4, we present a concise overview of the training process, including the number of hidden layers and the neuron count within each neural network layer. Furthermore, we evaluate the performance of the model across various driving scenarios and environmental temperatures. Finally, our conclusions are summarized in Section 5.

2. Description of the proposed temperature prediction algorithm

2.1. The theoretical basis of the proposed algorithm

Empirical mode decomposition (EMD) is an effective method for analyzing nonlinear and non-stationary signals [30]. This decomposition relies on the local temporal feature scale of the data to obtain intrinsic mode functions (IMFs), which highlight the inherent oscillatory modes of the original data across different time scales. One notable advantage of EMD is its lack of predetermined basis functions during decomposition, allowing it to adaptively capture both local oscillatory features and global trends present in the original sequence. The process of extracting IMF through EMD is referred to as decomposition. During this process, the original signal is decomposed into a series of IMFs, with each IMF representing vibration or variation patterns at different scales. This decomposition procedure forms the core of the EMD method, breaking down the signal into multiple IMFs, each satisfying specific local characteristics, enhancing the analysis and comprehension of the signal's oscillatory patterns. The detailed decomposition steps are illustrated in Fig. 1. By iteratively decomposing from high to low frequencies, multiple IMF components can be obtained. The time series can be represented through EMD decomposition as follows:

$$u(t) = \sum_{i=1}^{n} imf_i(t) + r(t) \tag{1}$$

where $imf_i(t)$ represents the intrinsic mode function of $u(t)$, and $r(t)$ represents the residual term.

When dealing with long-time series data, RNNs immediately come to mind. RNNs, which are a type of supervised machine learning model, are made of neurons with feedback loops [31]. In traditional RNNs, information can be propagated over time steps, but due to the continuous multiplication of gradients, it may lead to the vanishing or exploding gradients problem. This issue limits the capability of RNNs to handle long sequences. To solve these problems, the LSTM neural network is adopted. LSTM addresses the vanishing and exploding gradient problems by introducing three gating mechanisms: the input gate, forget gate, and output gate [32]. Each gate utilizes a sigmoid activation function to control the flow of information and selectively update or forget information through element-wise multiplication operations. The LSTM network model excels at learning long-term dependencies and

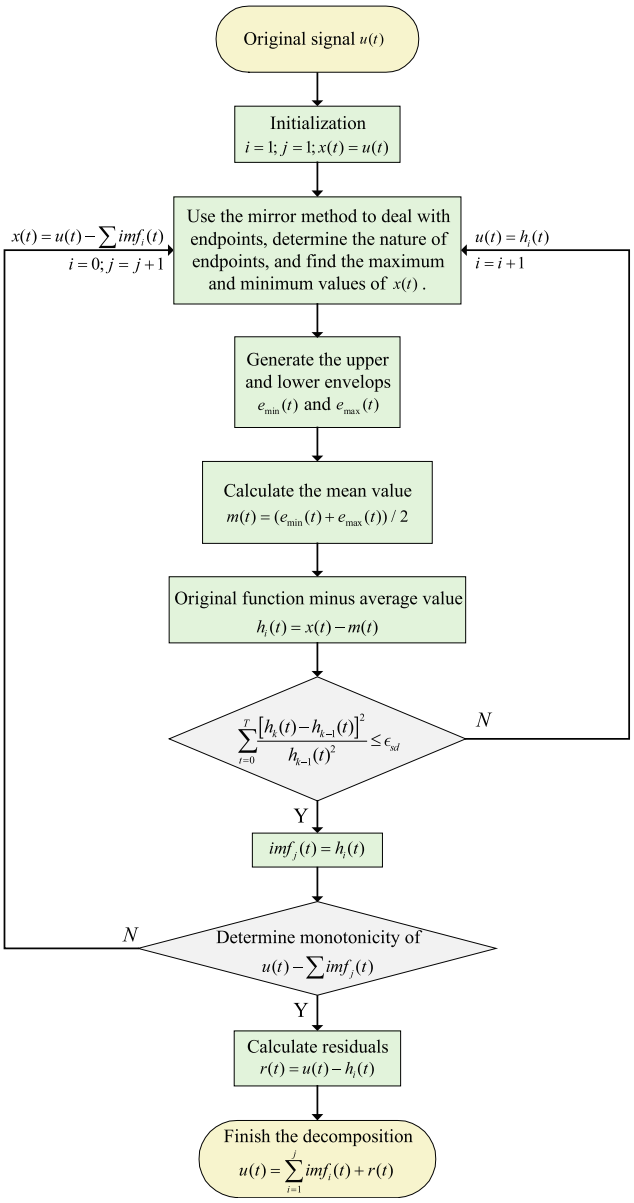


Fig. 1. EMD processing flow.

possesses a memory function by inferring the subsequent state from the preceding state, thus effectively enhancing the predictive accuracy of the model. The structure of the LSTM is shown in Fig. 2. Within the LSTM architecture, the forget gate governs the extent to which past memories are preserved, enabling a selective process of forgetting. The output of the forget gate is represented as follows:

$$f_t = \sigma\left(W_f \cdot \left[h_{t-1}, x_t\right] + b_f\right) \tag{2}$$

where σ represents the sigmoid activation function, W_f denotes the forget gate weight matrix, b_f represents the bias vector, h_{t-1} represents the previous time step output, and x_t represents the current time step input.

The input gate engages in memory selection, determining whether the current input x_t should be stored in the new cell state C_t. The output of the input gate is as follows:

$$i_t = \sigma\left(W_i \cdot \left[h_{t-1}, x_t\right] + b_i\right) \tag{3}$$

where W_i denotes the input gate weight matrix, and b_i represents the bias.

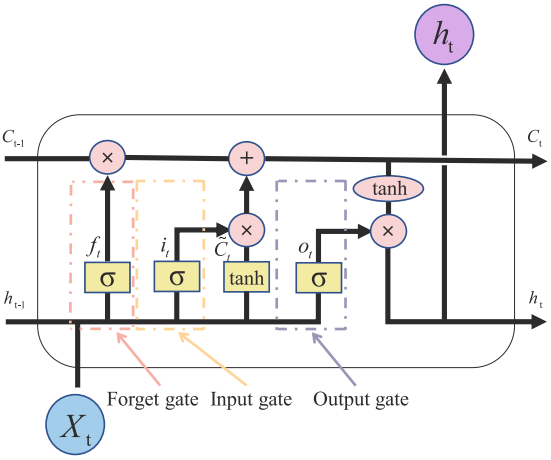


Fig. 2. The structure of the LSTM.

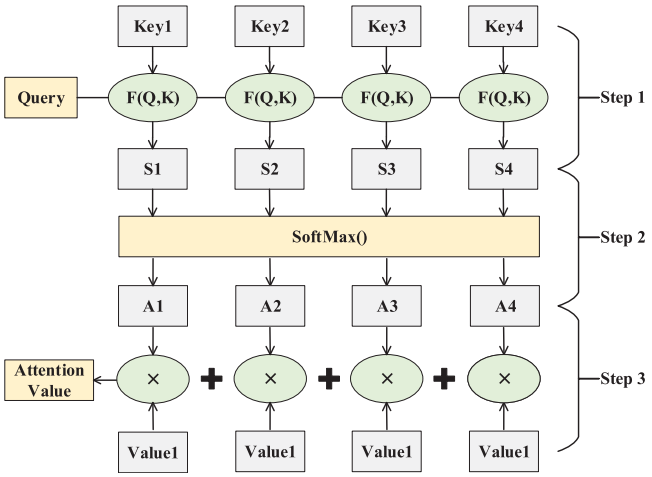


Fig. 3. AM process diagram.

The output i_t from the input gate combines with the output $\tilde{C}_t$ from the neighboring tanh layer to generate candidate information for updating and storing the memory cell state. Subsequently, this information interacts with the previous memory cell state C_{t-1} and the output of the forget gate, resulting in the creation of a new memory cell state C_t. The specific calculations are outlined below:

$$\tilde{C}_t = \tanh\left(W_C \cdot \left[h_{t-1}, x_t\right] + b_C\right) \tag{4}$$

$$C_t = f_t \otimes C_{t-1} + i_t \otimes \tilde{C}_t \tag{5}$$

where W_c represents the candidate input gate weight matrix, b_c represents the bias and $\otimes$ is the elementwise multiplication operator.

The output gate engages in the selection of outputs, determining which information from the new memory cell state C_t will be considered for the current state output. The output of the output gate is presented as follows:

$$o_t = \sigma\left(W_o\left[h_{t-1}, x_t\right] + b_o\right) \tag{6}$$

where W_o denotes the output gate weight matrix, and b_o represents the bias.

Finally, the output h_t of the LSTM at time step t is given by:

$$h_t = o_t \otimes \tanh\left(C_t\right) \tag{7}$$

To enhance the predictive accuracy of the model in handling long sequence data, this paper introduces an attention mechanism into the LSTM network. The concept of attention mechanism (AM) was introduced by Treisman and colleagues in 1980 [33]. This approach involves computing the probability distribution of attention to select crucial information from a vast amount of data, thereby emphasizing key inputs and optimizing traditional models. The core idea of the AM draws inspiration from the human visual attention process. Human vision rapidly identifies essential regions, focusing attention on them to gather necessary detailed information. Similarly, the attention mechanism selectively concentrates on more significant information, disregards less important details, and assigns varying degrees of importance to different pieces of information.

Structural models based on AM possess the capability to capture positional relationships among pieces of information, while also gauging the significance of different features based on their respective weights. By establishing dynamic weight parameters through discerning relevant and irrelevant information features, AM enhances crucial information and diminishes less useful details, thus elevating the efficiency of deep learning algorithms. Simultaneously, this approach addresses certain drawbacks present in traditional deep learning methods. AM has the capability to focus on important information with high weights while ignoring irrelevant details with low weights. Furthermore, it continuously adjusts these weights, allowing it to select crucial information even in varying circumstances. As a result, it exhibits higher scalability and robustness, making it well-suited for diverse scenarios [34].

The core formula of the attention mechanism is expressed:

$$\mathrm{AM}(Q, K, V) = \mathrm{Softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V \tag{8}$$

where Q, K, and V represent *Query*, *Key*, and *Value* respectively, and $\sqrt{d_k}$ denotes the dimensionality of K. The specific computational process, as illustrated in Fig. 3, can be divided into the following three stages:

Step 1. Attention scores are computed based on the similarity or correlation between the output features *Query* and *Key*. The attention score s_t is calculated as follows:

$$s_t = \tanh(W_h x_t + b_h) \tag{9}$$

where W_h represents the weight of the AM, b_h is the bias vector of the AM, and x_t denotes the input vector (W_h and b_h are determined by *Query*).

Step 2. In this phase, the attention scores obtained from the first stage are normalized to form a weight distribution within the range of [0, 1]. The normalized attention scores a_i is calculated as follows:

$$a_i = \frac{\exp\left(s_i\right)}{\sum_{j=1}^{L}\exp\left(s_j\right)} \tag{10}$$

Where L represents the length of the sequence.

Step 3. By weighting the values according to the weight coefficients, the final attention score is obtained. The final attention scores $s_{attention}$ is calculated as follows:

$$s_{attention} = \sum_t a_t h_t \tag{11}$$

where a_t represents the corresponding score weight, and h_t represents the *value* determined by the input sequence.

This paper combines LSTM with an AM to preserve temporal information while reducing network parameters, mitigating computational load, and enhancing processing speed. The LSTM-AM structure introduces an additional fully connected layer between the hidden layer and the output layer of the original LSTM network. Through this layer's operations, a weighted combination is applied to the outputs of multiple hidden neurons across various time steps. Consequently,

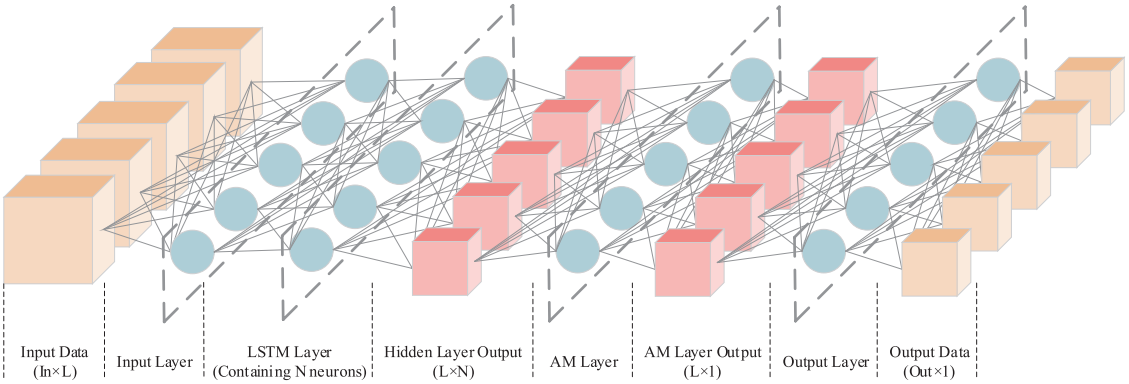


Fig. 4. Architecture of the LSTM-AM model.

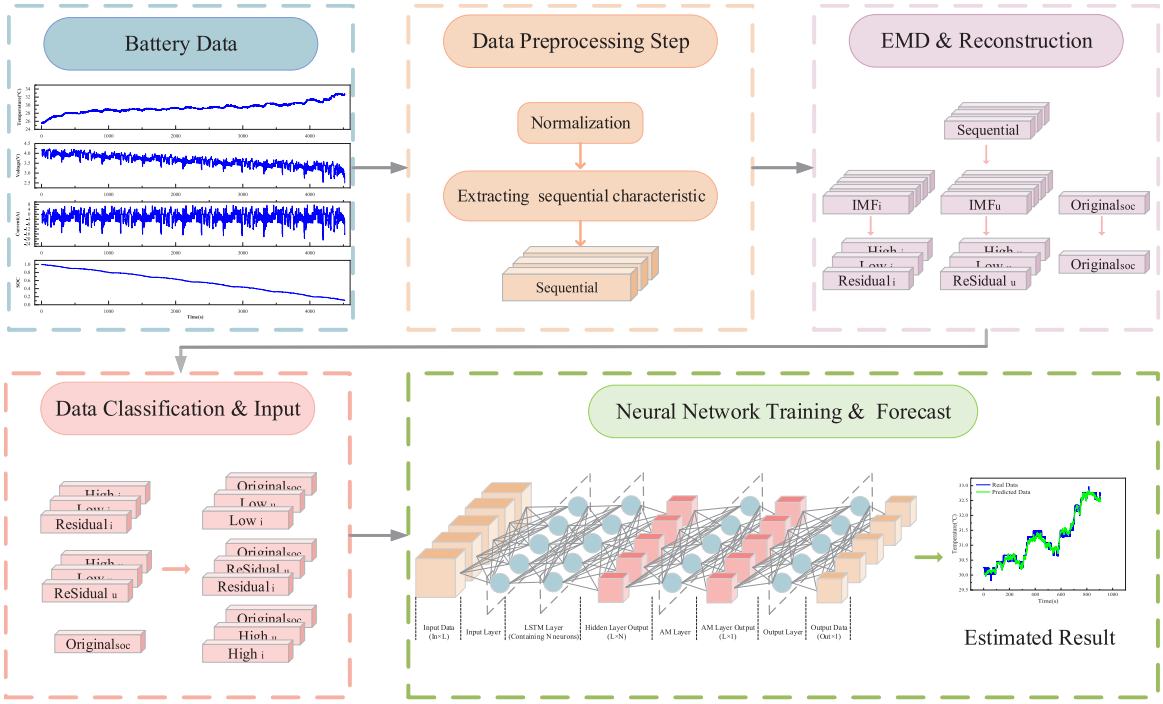


Fig. 5. The architecture of the proposed temperature prediction algorithm.

the dimensionalities of the output data can be reduced to $L \times 1$ before connecting with the output layer. Importantly, the connection weights between the output values at each time step and the attention mechanism layer are shared.

As illustrated in Fig. 4, assuming the number of neurons in the attention layer is N_a, the required parameter count at this point is $(N_h+1)\times N_a+(L+1)\times N_a$. The introduction of the attention mechanism substantially diminishes the influence of the number of hidden layer neurons on the parameter count. This is advantageous for improving the precision and computational efficiency of the model.

2.2. Analysis of structural characteristics in battery temperature data

The original temperature data of the battery undergoes EMD, resulting in multiple IMFs and one residual component. IMFs represent the natural oscillation modes of the data signal, each characterized by distinct frequency features. Generally, high-frequency components signify short-term fluctuations or noise within the battery temperature data. They are often induced by measurement errors, environmental interference, or rapid responses from the battery itself. Low-frequency components typically reflect the primary patterns of battery temperature variations, such as charge–discharge cycles and long-term trends in temperature fluctuations. The residual component represents the overall trend of the battery temperature and can be influenced by factors like battery aging, and changes in environmental temperature.

Following the mentioned patterns, this paper categorizes the IMF sequence into high and low frequencies. IMF_1 is denoted as $Indicator_1$, the sum of IMF_1 and IMF_2 as $Indicator_2$, and so forth. The sum of IMF_1 to IMF_i as $Indicator_i$. The mean of $Indicator_1$ to $Indicator_n$ is calculated and a t-test is conducted to determine whether this mean significantly differs from 0. The t-test statistic is given by:

$$t = \left(\bar{X}_i - 0\right) / \frac{\sigma_i}{\sqrt{n-1}} \tag{12}$$

where $\bar{X}_i$ represents the mean of Indicator i, σ_i denotes the standard deviation of the indicator, and n signifies the sample size of the indicator. If there is a significant difference in the mean value of the IMF at $Indicator_a$, then IMF_1, IMF_2, …, IMF_a represent high-frequency components, while IMF_{a+1}, IMF_{a+2}, …, IMF_n serve as low-frequency components.

2.3. The architecture of the proposed algorithm

This paper introduces a lithium-ion battery surface temperature prediction method named EMD-LSTM-AM, which is based on frequency domain decomposition. The architecture of this method is illustrated in Fig. 5, and the main steps are outlined as follows:

Step 1. Normalize the input data and divide it into temporal sequences with a predefined time step.
Step 2. Decompose the temporal sequences into several IMF components through EMD.
Step 3. Reconstruct the IMFs components into high-frequency, low-frequency, and trend components through frequency domain decomposition.
Step 4. Reorganize features based on different frequencies and use them as inputs for three separate subnetworks.
Step 5. The reconstructed data are inputted into the trained LSTM-AM to get the corresponding output value.
Step 6. The outputs from the sub-networks are accumulated to obtain the final predictive output.

$$T_{\text{estimated}} = \sum T_{\text{Subnetwork}_i} \tag{13}$$

where $T_{\text{estimated}}$ represents the final predicted temperature and $T_{\text{Subnetworki}}$ denotes the output of subnetwork i.

3. Experimental data and evaluation indicator

3.1. Experimental data

The data used in this study was sourced from a publicly available dataset provided by the University of Wisconsin–Madison [35]. The experiments on this dataset were conducted within an 8-cubic-foot environmental chamber, employing a Digatron Firing Circuits universal battery testing apparatus rated at 25 A and 18 V. Throughout the testing process, pristine 2.9Ah Panasonic 18650PF cells were employed. Data collection occurred at intervals of 0.1 s, with corresponding records being documented.

During the experiment, the battery underwent a complete charging cycle employing the constant-current constant-voltage (CCCV) technique. Subsequently, it was discharged following a combination of diverse driving profiles, including the urban dynamometer driving schedule (UDDS), Los Angeles 92 (LA92), supplemental federal test procedure driving schedule (US06), and the highway fuel economy driving schedule (HWFET). Fig. 6 depicts the curves of battery current, battery terminal voltage, SOC, and battery surface temperature (ST) under the US06 drive cycle at 25 °C.

3.2. Evaluation criteria

To quantitatively assess performance, we employed three established evaluation criteria: root mean square error (RMSE), mean absolute error (MAE), and maximum error (MAXE), formulated as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\hat{y}_i - y_i\right)^2} \tag{14}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|\hat{y}_i - y_i\right| \tag{15}$$

$$MAXE = Max\left|\hat{y}_i - y_i\right| \tag{16}$$

where n represents the length of the dataset, $\hat{y}_i$ signifies the predicted value, and y_i denotes the actual value. The RMSE reflects the robustness of the prediction, while MAE and MAXE quantify the accuracy of the prediction.

4. Experimental analyses

This chapter provides a detailed exploration of the effectiveness and analysis of the algorithm proposed in this paper under various environmental conditions and diverse driving scenarios with varying temperatures. Furthermore, it outlines the model hyperparameter settings and the training process. The test data for this chapter were sourced from the University of Wisconsin–Madison. The dataset was divided into two parts for testing purposes, with the first 80% allocated for training, while the remaining 20% was reserved for validation.

4.1. Optimization of neural network parameters

Since the potential accuracy of a neural network model depends on its parameter configuration, it is crucial to explore various network settings to achieve the highest prediction precision. Key parameters for the proposed model in this paper include the number of layers and the count of hidden nodes in each layer. Advanced configurations encompass aspects such as batch size, inter-layer dropout rates, and the learning rate used during model training.

Some of the parameters during the training process are initialized as follows:

(1) Training epoch: 1000 times.
(2) Mini-batch size: 256.
(3) Dropout rare: 20%.
(4) Optimizer: Adam.

The proposed model in this paper consists of three sub-models, with the size of the input sequence layer and the output layer matching the dimensions of the input vector and output data, respectively, set at 3 and 1. Within the model, the number of hidden neurons governs its width, while the depth is determined by the number of hidden layers. These two values are critical hyperparameters in the model and require meticulous tuning. The experiments employed multiple consecutive powers of 2 to determine the count of hidden neurons [36]. The training outcomes using this approach are presented in Table 1.

As shown in Table 1, within Subnetwork 1, when the LSTM-AM has only one hidden layer, the RMSE decreases from 0.0445 °C for 4 nodes to 1.0282 °C for 128 nodes. Considering MAE and MAXE, within the single hidden layer structure, networks with 4 and 8 nodes exhibit the highest accuracy. A multi-layer LSTM-AM model is constructed by stacking the 4-node and 8-node networks from the first layer. The evaluation continues by computing three metrics to assess its performance. Sub-Networks 2 and 3 undergo a similar training process. The final results indicate that Sub-Network 1 achieves excellent data fitting with just one hidden layer and 4 hidden neurons. For Sub-Networks 2 and 3 to achieve optimal prediction performance, two hidden layers with 16 hidden neurons each are required. Fig. 7 depicts the evaluation error indexes and outcomes of the optimal architecture.

The above experiments demonstrate that a stacked LSTM-AM network with a small number of hidden layers and neurons can perform reasonably accurate surface temperature prediction for lithium-ion batteries. This can be attributed to the low dimensionality of the input vectors in the data, making a small number of hidden neurons sufficient to capture the nonlinear characteristics within the lithium-ion battery. Models with an excessive number of hidden neurons often tend to overfit, resulting in biased predictive performance.

4.2. Analysis of battery surface temperature prediction under different methods

In this section, to validate the performance of the model proposed in this paper, experiments were conducted by comparing it with traditional RNN, gate recurrent unit (GRU), LSTM, EMD-LSTM, and LSTM-AM models using the same dataset. At the same time, to

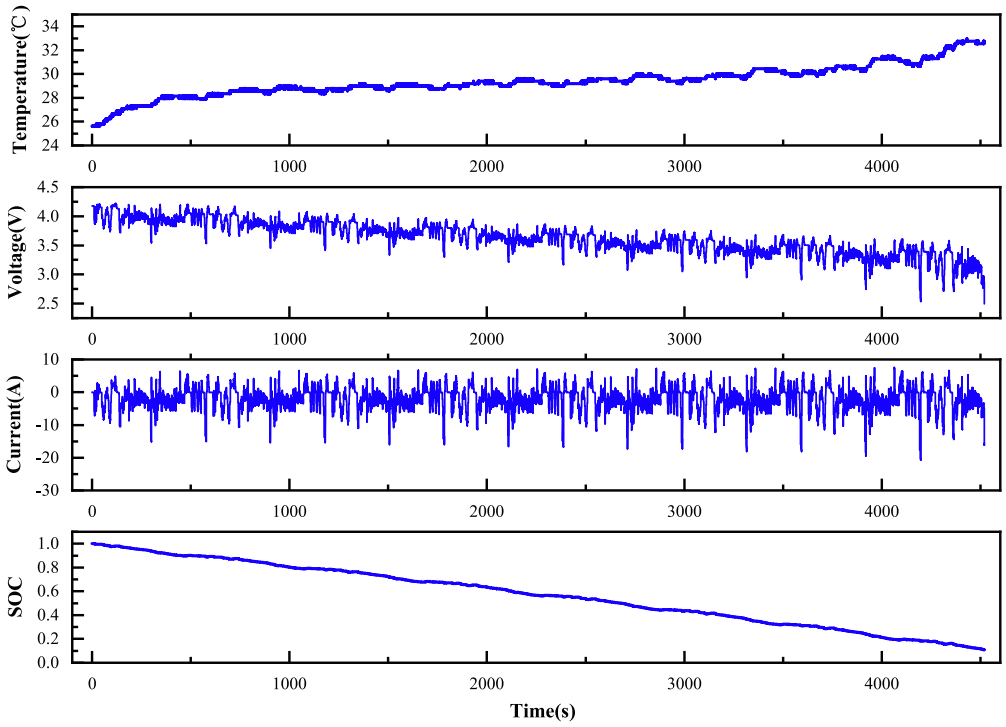


Fig. 6. US06 drive cycle under 25 °C: temperature, voltage, current and SOC.

Table 1
Performance metrics of networks with different parameters.

Hidden neurons	Number	Layer=1						Layer=2				Layer=3		
		4	8	16	32	64	128	4	8	16	32	8	16	32
Residual data (Subnetwork 1)	RMSE (°C)	0.0445	0.0476	0.5291	0.8701	0.5601	1.0282	0.0823	0.2253	0.7386	3.2546	0.8840	0.8820	1.0261
	MAE (°C)	0.0363	0.0388	0.4144	0.7000	0.4668	0.8639	0.0679	0.2050	0.6002	3.0062	0.7368	0.7377	0.8590
	MAXE (°C)	0.0867	0.0850	1.2583	1.4839	0.9470	1.6540	0.2441	0.3269	1.2795	5.7614	1.5066	1.4583	1.7829
Low-frequency data (Subnetwork 2)	RMSE (°C)	0.1374	0.1110	0.0744	0.1055	0.1071	0.1230	0.0829	0.0808	0.0596	0.0865	0.1277	0.0872	0.1439
	MAE (°C)	0.1067	0.0813	0.0640	0.0796	0.0783	0.0874	0.0617	0.0526	0.0472	0.0577	0.0981	0.0649	0.0924
	MAXE (°C)	0.4353	0.3507	0.1789	0.4630	0.5792	0.4371	0.2364	0.3501	0.1950	0.3961	0.3799	0.3424	0.5394
High-frequency data (Subnetwork 3)	RMSE (°C)	0.0868	0.0865	0.0857	0.0859	0.0859	0.0861	0.0847	0.0867	0.0594	0.0861	0.0869	0.0868	0.0866
	MAE (°C)	0.0674	0.0674	0.0672	0.0667	0.0668	0.0671	0.0670	0.0673	0.0443	0.0669	0.0680	0.0682	0.0675
	MAXE (°C)	0.2918	0.2977	0.2875	0.2898	0.2993	0.2942	0.2709	0.2901	0.2150	0.2718	0.3058	0.3421	0.2995

Table 2
Comparison of evaluation error indexes of the five methods.

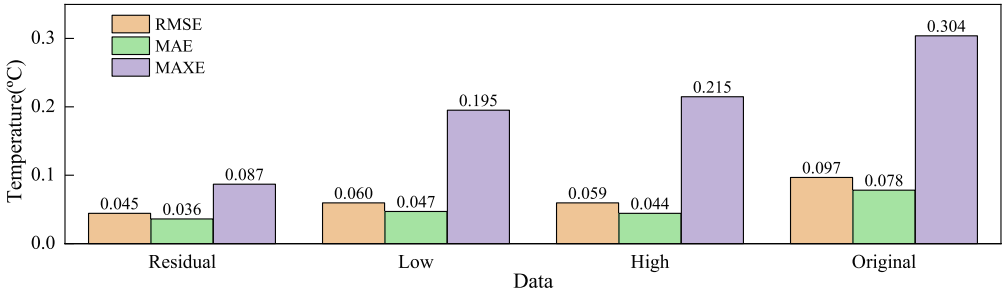
Method	RNN	GRU	LSTM	EMD-LSTM	LSTM-AM	**EMD-LSTM-AM**
RMSE	0.3476	0.1101	0.1108	0.1033	0.1000	**0.0980**
MAE	0.2747	0.0869	0.0829	0.0789	0.0824	**0.0784**
MAXE	0.9178	0.4340	0.5350	0.4073	0.3400	**0.3043**
Parameter count	2081	3601	4353	9075	4625	9639

assess the complexity of the model, we quantify this metric by measuring the trainable parameter count. All methods were implemented in Python using the open-source deep learning library Keras based on the TensorFlow framework.

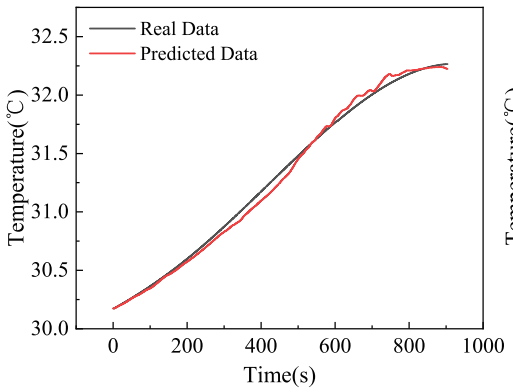
As shown in Fig. 8, among the five prediction methods, the degree of line fitting between actual values and predicted values, ranked from highest to lowest, is as follows: EMD-LSTM-AM, LSTM-AM, EMD-LSTM, LSTM, GRU, and RNN. EMD-LSTM-AM exhibits the highest degree of line fitting, with real and predicted values nearly perfectly aligned, while RNN has the lowest degree of line fitting. By comparing the predicted values obtained from different methods with the actual values, various evaluation error metrics for each method can be calculated.

As shown in Table 2, compared to RNN, LSTM shows improvements in all evaluation metrics, with a reduction of 0.23 °C in RMSE, a decrease of 0.19 °C in MAE, and a lowering of 0.37 °C in the maximum error. Compared to the RNN algorithm, the introduction of gate structures in LSTM, which enable control over the flow of information and the ability to determine what to retain or forget, renders it more appropriate for temperature prediction of batteries, particularly in the context of handling sequential data.

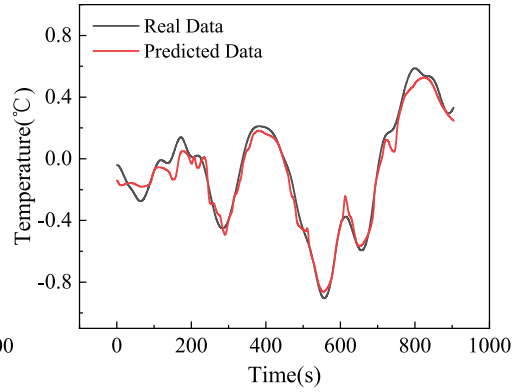
Comparing LSTM to GRU, RMSE is comparable, while MAE shows a slight improvement. LSTM networks exhibit stronger long-term memory capabilities, allowing them to retain more historical information, thereby enhancing the effectiveness of the attention mechanism. While GRU offers a simpler and more computationally efficient structure, it might lose some long-range information in certain scenarios, leading to a decrease in the performance of the attention mechanism. Therefore, in situations with similar metrics, this paper opts for LSTM networks. For LSTM-AM in comparison to LSTM, it shows a reduction of 0.0108 °C in RMSE, a decrease of 0.005 °C in MAE, and a lowering of 0.195 °C in MAXE. These results indicate that, for LSTM, the introduction of



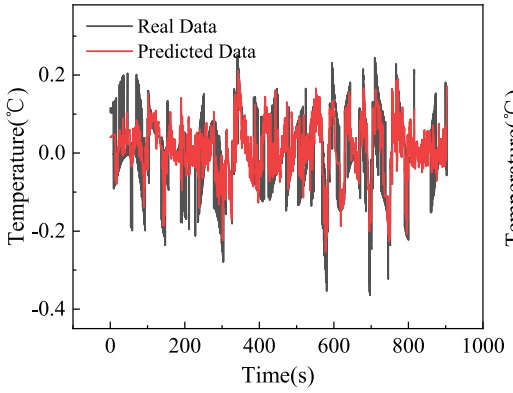
(a) Evaluation error indexes for the three sub-networks and final outcome.



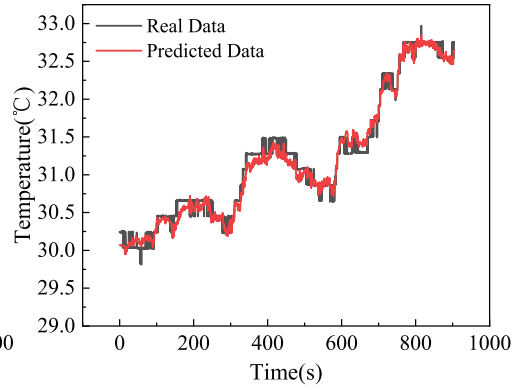
(b) residual data.



(c) low-frequency data.

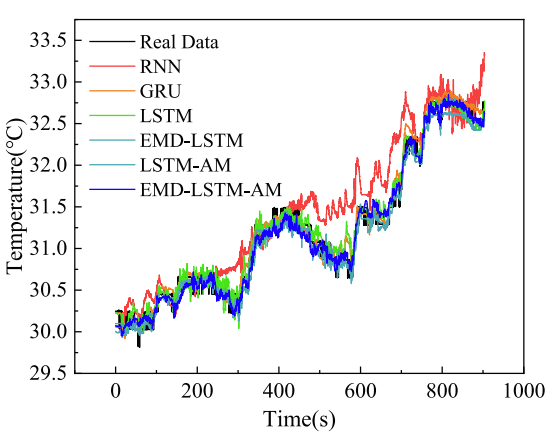


(d) high-frequency data.

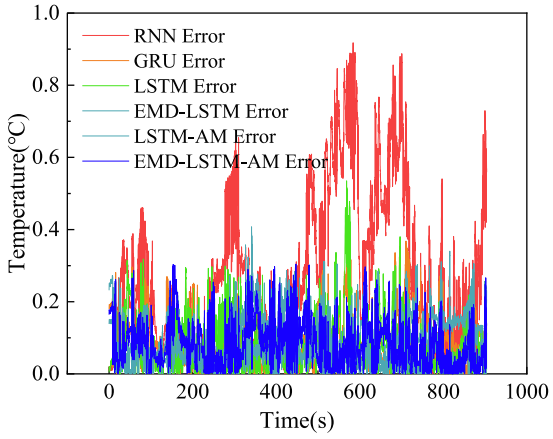


(e) original data.

Fig. 7. Temperature prediction using the LSTM-AM network with the proposed method under US06 loading profiles and 25 °C.

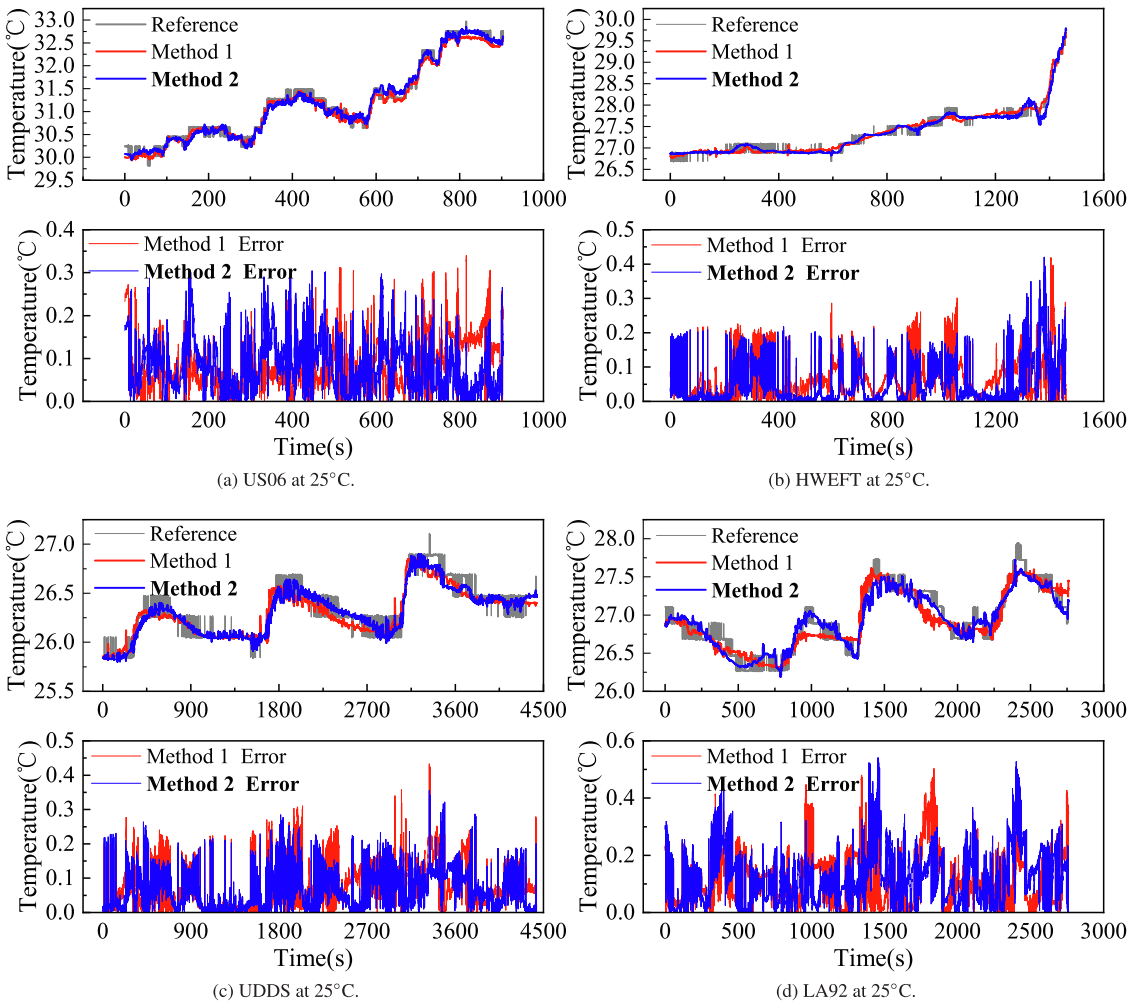


(a) Comparison of temperature predictions.



(b) Absolute error.

Fig. 8. Predictive performance and absolute errors of Six different methods.



(a) US06 at 25°C. (b) HWEFT at 25°C.

(c) UDDS at 25°C. (d) LA92 at 25°C.

Fig. 9. Temperature prediction using the LSTM-AM network with the proposed method under different loading profiles and 25 °C. (Method 1 is the LSTM-AM network, while Method 2 is the EMD-LSTM-AM network.)

an attention mechanism can effectively improve predictive accuracy. The EMD-LSTM architecture introduces EMD on top of the LSTM. This addition results in slight improvements in both RMSE and MAE metrics, while a more substantial improvement is observed in the MAXE metric. Therefore, employing EMD decomposition and reconstruction in input features of LSTM-AM can further enhance prediction accuracy. On the other hand, while the incorporation of EMD and AM increases the complexity of the model, the proposed network consists of three distinct subnetworks. Since these three sub-networks can run in parallel, the overall computational speed of the network is not significantly affected. Therefore, we consider the increase in complexity to be reasonable, as it ensures computational speed while enhancing prediction accuracy.

Experimental results demonstrate that among the six methods, EMD-LSTM-AM exhibits the best performance. Therefore, the EMD-LSTM-AM method proposed in this paper can provide the most accurate prediction of the battery surface temperature for the next time step. It effectively adapts to changes in high-frequency noise and non-stationary noise, thereby offering a high-precision and high-frequency reference for battery management systems.

4.3. Prediction and analysis of battery surface temperature under various driving conditions

The temperature of automotive batteries is influenced by various factors, including the type of battery, the composition of the electrolyte, the rate of charge and discharge, and the external environmental temperature. Different driving cycles result in varying charge–discharge frequencies and intensities, which in turn affect the internal chemical reactions and heat conduction within the battery, subsequently impacting its temperature. In this section, data from various driving conditions are employed to validate the effectiveness of the proposed approach. Additionally, a comparison is made between the proposed network and the LSTM-AM network. Method 1 represents the LSTM-AM network, while method 2 corresponds to the method introduced in this paper. Fig. 9 and Table 3 respectively present the evaluation metrics, predicted temperatures, and prediction errors of the model under four different operating conditions.

The results in Table 3 indicate that the proposed network achieves the lowest RMSE and MAE under the HWFET condition. This is because the HWFET condition is relatively smooth, with no frequent starts and stops or rapid accelerations and decelerations. The highest vehicle speed during its driving cycle is 96.6 km/h, and the average speed is 77.7 km/h, which is closer to the mean. Consequently, the battery temperature changes more gradually, making it easier for the model to identify its internal nonlinear characteristics.

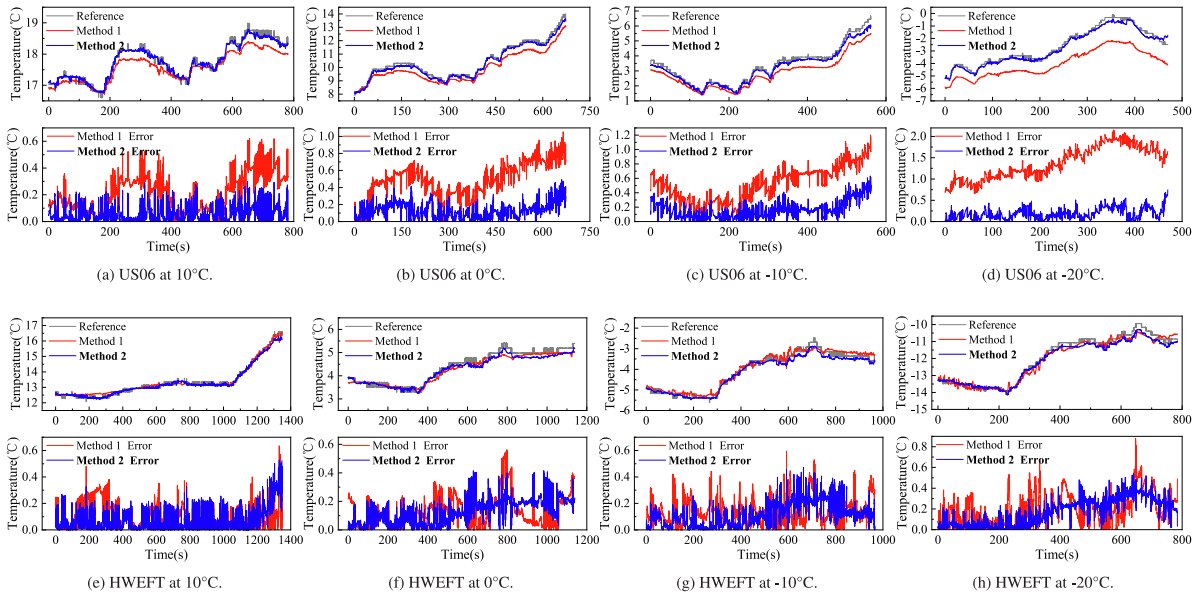
On the other hand, the model exhibits poorer performance under the US06 and LA92 conditions. This can be attributed to the aggressive nature of the US06 driving cycle, which involves significant accelerations, decelerations, and gear shifts. Similarly, the LA92 driving cycle is relatively complex, featuring multiple rapid accelerations and decelerations. In the LSTM network, when input feature data undergo frequent and significant changes, it can become challenging for LSTM

Table 3
Comparison of evaluation error indexes of the different drive cycles.

Drive cycles	US06			HWFET			UDDS			LA92		
Model	Method 1	**Method 2**	Improvement	Method 1	**Method 2**	Improvement	Method 1	**Method 2**	Improvement	Method 1	**Method 2**	Improvement
RMSE	0.1000	**0.0980**	2.00%	0.0884	**0.0734**	16.89%	0.0976	**0.0766**	21.44%	0.1494	**0.1198**	19.77%
MAE	0.0824	**0.0784**	4.86%	0.0657	**0.0465**	29.30%	0.0743	**0.0580**	21.92%	0.1168	**0.0939**	19.65%
MAXE	0.3400	**0.3043**	10.49%	0.4223	**0.4200**	0.55%	0.4332	**0.3559**	17.86%	0.5030	**0.4743**	5.69%

Table 4
Comparison of evaluation error indexes of the different temperature conditions.

	Drive cycles	US06			HWFET		
	Model	Method 1	**Method 2**	Improvement	Method 1	**Method 2**	Improvement
25 °C	RMSE (°C)	0.1000	**0.0980**	2.00%	0.0884	**0.0734**	16.89%
	MAE (°C)	0.0824	**0.0784**	4.86%	0.0657	**0.0465**	29.30%
	MAXE (°C)	0.3400	**0.3043**	10.49%	0.4223	**0.4200**	0.55%
10 °C	RMSE (°C)	0.2522	**0.0763**	69.73%	0.1334	**0.1025**	23.12%
	MAE (°C)	0.2122	**0.0549**	74.11%	0.0996	**0.0651**	34.61%
	MAXE (°C)	0.6206	**0.3012**	51.46%	0.6443	**0.5271**	18.20%
0 °C	RMSE (°C)	0.5238	**0.1533**	70.74%	0.1703	**0.1504**	11.73%
	MAE (°C)	0.4805	**0.1264**	73.69%	0.1329	**0.1234**	7.12%
	MAXE (°C)	1.0504	**0.4929**	53.08%	0.5616	**0.4145**	26.20%
−10 °C	RMSE (°C)	0.2669	**0.1981**	25.79%	0.1704	**0.1448**	15.04%
	MAE (°C)	0.2344	**0.1593**	32.06%	0.1416	**0.1157**	18.28%
	MAXE (°C)	0.7791	**0.6323**	18.85%	0.5982	**0.4705**	21.35%
−20 °C	RMSE (°C)	1.4329	**1.1125**	22.36%	0.2423	**0.2109**	12.93%
	MAE (°C)	1.3824	**1.1123**	19.54%	0.1974	**0.1687**	14.55%
	MAXE (°C)	2.1527	**1.1409**	47.00%	0.8784	**0.5470**	37.72%



(a) US06 at 10°C. (b) US06 at 0°C. (c) US06 at -10°C. (d) US06 at -20°C.

(e) HWEFT at 10°C. (f) HWEFT at 0°C. (g) HWEFT at -10°C. (h) HWEFT at -20°C.

Fig. 10. Temperature estimation using the LSTM-AM network with the proposed method under different temperature conditions. (Method 1 is the LSTM-AM network, while Method 2 is the EMD-LSTM-AM network.)

to capture long-term dependencies within the sequence. While LSTM can mitigate the vanishing gradient problem, it may still be susceptible to issues like exploding gradients or overfitting. If input feature data undergoes too frequent or drastic changes, LSTM may struggle to adapt to these variations, thus reducing its generalization capacity and predictive accuracy.

4.4. Prediction and analysis of battery surface temperature under different temperature conditions

Because environmental temperature significantly affects the prediction of battery surface temperature, this section validates the proposed network under various fixed environmental temperature conditions and compares it with the LSTM-AM network. Similar to Section 4.3, method 1 represents the LSTM-AM network, while method 2 corresponds to the method introduced in this paper. Fig. 5 and Table 4 presented below depict the measured temperature, predicted temperature, prediction errors, and model evaluation metrics under environmental temperature conditions of −20 °C, −10 °C, 0 °C, 10 °C, and 25 °C for both the US06 and HWFET driving cycles.

The results in Fig. 10 and Table 4 suggest that the proposed network exhibits higher estimation errors under low-temperature conditions. This outcome indicates that the proposed algorithm performs less effectively at low temperatures. The reason for this lies in the fact that low temperatures can significantly impact the dynamic characteristics of lithium-ion batteries. This results in a reduction in the diffusion rate of lithium ions within the electrode, leading to gradients in lithium

concentration and phase changes within the electrode. These factors can cause non-uniformity in current density and overpotentials at the electrode surface, which subsequently affects the deposition and dissolution of lithium, resulting in reduced uniformity in battery surface temperature. This, in turn, impacts the learning and generalization capabilities of the neural network.

Compared to the LSTM-AM network, the method proposed in this paper demonstrates superior performance, especially in cold environments where the contrast is more pronounced. This is attributed to the introduction of EMD, which enables better recognition of temperature trends and reduces the impact of physical property changes in batteries at low temperatures on the network.

5. Conclusion

This paper proposes a frequency reconstruction oriented EMD-LSTM-AM based surface temperature prediction method for lithium-ion batteries. The effectiveness of the proposed method is verified by comparing it with six different neural network architectures. Furthermore, the generalization and stability of the model are verified under five different temperature conditions and four driving cycles. At 25 °C under US06 conditions, the maximum error predicted by the model is less than 0.31 °C, and the RMSE is less than 0.1 °C. Even in low-temperature environments (−20 °C), the maximum error is only 1.14 °C. The comparative results in the paper demonstrate that the proposed method outperforms other popular prediction methods across various metrics. This data-driven model relies solely on temperature and current for self-learning, resulting in low computational requirements after training and lower prediction temperature errors, demonstrating robustness. Therefore, it only requires real-time collection of current and other inputs to predict the battery surface temperature.

In summary, the frequency reconstruction oriented EMD-LSTM-AM based surface temperature prediction method for lithium-ion batteries exhibits high prediction accuracy and strong reliability. It can effectively adapt to the impact of high-frequency and non-stationary noise caused by high prediction frequencies, making it promising for real-world vehicle temperature prediction. However, due to potential battery aging issues that may result in parameter changes, it becomes necessary to retrain the model to meet the performance requirements of aging batteries. Integrating physical models for real-time online updating of neural network weights will be a focal point of our future work.

CRediT authorship contribution statement

Xiao Qi: Formal analysis, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing. **Chaofeng Hong:** Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Tao Ye:** Methodology, Visualization. **Lijun Gu:** Supervision, Visualization, Writing – review & editing. **Weixiong Wu:** Conceptualization, Investigation, Project administration, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests orpersonal relationships that could have appeared to influence the work reportedin this paper.

Data availability

Data will be made available on request.

Acknowledgments

This work is supported by the National Natural Science Foundation of China (No. 52106244), Guangdong Basic and Applied Basic Research Foundation (No. 2021A1515110016, No. 2022A1515011936), and Science and Technology Project of China Southern Power Grid under Grant GDKJXM20230246(030100KC23020017).

References

[1] B.-C. Wang, Y. Feng, S. Wang, Two-phase separation based spatiotemporal modeling of thermal processes with applications to lithium-ion batteries, J. Energy Storage 49 (2022) 104050.
[2] F. Xiao, C. Li, Y. Fan, G. Yang, X. Tang, State of charge estimation for lithium-ion battery based on Gaussian process regression with deep recurrent kernel, Int. J. Electr. Power Energy Syst. 124 (2021) 106369.
[3] X. Zhang, Z. Li, L. Luo, Y. Fan, Z. Du, A review on thermal management of lithium-ion batteries for electric vehicles, Energy 238 (2022) 121652.
[4] S. Chen, G. Zhang, C. Wu, W. Huang, C. Xu, C. Jin, Y. Wu, Z. Jiang, H. Dai, X. Feng, et al., Multi-objective optimization design for a double-direction liquid heating system-based Cell-to-Chassis battery module, Int. J. Heat Mass Transfer 183 (2022) 122184.
[5] Y. Feng, L. Zhou, H. Ma, Z. Wu, Q. Zhao, H. Li, K. Zhang, J. Chen, Challenges and advances in wide-temperature rechargeable lithium batteries, Energy Environ. Sci. 15 (5) (2022) 1711–1759.
[6] X. Hu, L. Xu, X. Lin, M. Pecht, Battery lifetime prognostics, Joule 4 (2) (2020) 310–346.
[7] R.K. Daniels, V. Kumar, S.S. Chouhan, A. Prabhakar, Thermal runaway fault prediction in air-cooled lithium-ion battery modules using machine learning through temperature sensors placement optimization, Appl. Energy 355 (2024) 122352.
[8] G. Zhu, C. Kong, J.V. Wang, J. Kang, Q. Wang, C. Qian, A fractional-order electrochemical lithium-ion batteries model considering electrolyte polarization and aging mechanism for state of health estimation, J. Energy Storage 72 (2023) 108649.
[9] T. Wu, C. Wang, Y. Hu, Z. Liang, C. Fan, Research on electrochemical characteristics and heat generating properties of power battery based on multi-time scales, Energy 265 (2023) 126416.
[10] J. Wen, X. Chen, X. Li, Y. Li, SOH prediction of lithium battery based on IC curve feature and BP neural network, Energy 261 (2022) 125234.
[11] G. Lee, D. Kwon, C. Lee, A convolutional neural network model for SOH estimation of Li-ion batteries with physical interpretability, Mech. Syst. Signal Process. 188 (2023) 110004.
[12] J. Liu, S. Yadav, M. Salman, S. Chavan, S.C. Kim, Review of thermal coupled battery models and parameter identification for lithium-ion battery heat generation in EV battery thermal management system, Int. J. Heat Mass Transfer 218 (2024) 124748.
[13] W. Li, J. Zhu, Y. Xia, M.B. Gorji, T. Wierzbicki, Data-driven safety envelope of lithium-ion batteries for electric vehicles, Joule 3 (11) (2019) 2703–2715.
[14] U. Krewer, F. Röder, E. Harinath, R.D. Braatz, B. Bedürftig, R. Findeisen, Dynamic models of Li-ion batteries for diagnosis and operation: a review and perspective, J. Electrochem. Soc. 165 (16) (2018) A3656.
[15] W. Zhang, T. Li, W. Wu, N. Ouyang, G. Huang, Data-driven state of health estimation in retired battery based on low and medium-frequency electrochemical impedance spectroscopy, Measurement 211 (2023) 112597.
[16] N. Shi, Z. Chen, M. Niu, Z. He, Y. Wang, J. Cui, State-of-charge estimation for the lithium-ion battery based on adaptive extended Kalman filter using improved parameter identification, J. Energy Storage 45 (2022) 103518.
[17] S. Zhang, H. Zhu, J. Wu, Z. Chen, Voltage relaxation-based state-of-health estimation of lithium-ion batteries using convolutional neural networks and transfer learning, J. Energy Storage 73 (2023) 108579.
[18] J.Á. Antón, P.G. Nieto, F. de Cos Juez, F.S. Lasheras, M.G. Vega, M.R. Gutiérrez, Battery state-of-charge estimator using the SVM technique, Appl. Math. Model. 37 (9) (2013) 6244–6253.
[19] Y. Guo, Z. Zhao, L. Huang, Soc estimation of lithium battery based on improved BP neural network, Energy Procedia 105 (2017) 4153–4158.
[20] E. Chemali, P.J. Kollmeyer, M. Preindl, R. Ahmed, A. Emadi, Long short-term memory networks for accurate state-of-charge estimation of Li-ion batteries, IEEE Trans. Ind. Electron. 65 (8) (2017) 6730–6739.
[21] K. Fang, D. Mu, S. Chen, B. Wu, F. Wu, A prediction model based on artificial neural network for surface temperature simulation of nickel–metal hydride battery during charging, J. Power Sources 208 (2012) 378–382.
[22] Z. Zhao, Z. Chunjie, G. Xiangfeng, S. Hutang, Solar cell temperature prediction model of support vector machine optimized by particle swarm optimization algorithm, Acta Phys. Sin. 64 (8) (2015) 380–386.
[23] A.A. Hussein, A.A. Chehade, Robust artificial neural network-based models for accurate surface temperature estimation of batteries, IEEE Trans. Ind. Appl. 56 (5) (2020) 5269–5278.

[24] F. Yang, W. Li, C. Li, Q. Miao, State-of-charge estimation of lithium-ion batteries based on gated recurrent neural network, Energy 175 (2019) 66–75.
[25] S. Zhu, C. He, N. Zhao, J. Sha, Data-driven analysis on thermal effects and temperature changes of lithium-ion battery, J. Power Sources 482 (2021) 228983.
[26] J. Xie, R. Yang, H.B. Gooi, H.D. Nguyen, PID-based CNN-LSTM for accuracy-boosted virtual sensor in battery thermal management system, Appl. Energy 331 (2023) 120424.
[27] Y. Wang, X. Chen, C. Li, Y. Yu, G. Zhou, C. Wang, W. Zhao, Temperature prediction of lithium-ion battery based on artificial neural network model, Appl. Therm. Eng. (2023) 120482.
[28] W. Liu, X. Hu, X. Lin, X.-G. Yang, Z. Song, A.M. Foley, J. Couture, Toward high-accuracy and high-efficiency battery electrothermal modeling: A general approach to tackling modeling errors, Etransportation 14 (2022) 100195.
[29] N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. Hamprecht, Y. Bengio, A. Courville, On the spectral bias of neural networks, in: International Conference on Machine Learning, PMLR, 2019, pp. 5301–5310.
[30] N.E. Huang, Z. Shen, S.R. Long, M.C. Wu, H.H. Shih, Q. Zheng, N.-C. Yen, C.C. Tung, H.H. Liu, The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis, Proc. R. Soc. Lond. Ser. A Math. Phys. Eng. Sci. 454 (1971) (1998) 903–995.
[31] M. Kubat, Neural networks: a comprehensive foundation by simon haykin, Knowl. Eng. Rev. 13 (4) (1999) 409–412.
[32] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780.
[33] A.M. Treisman, G. Gelade, A feature-integration theory of attention, Cogn. Psychol. 12 (1) (1980) 97–136.
[34] S. Hao, D.-H. Lee, D. Zhao, Sequence to sequence learning with attention mechanism for short-term passenger flow prediction in large-scale metro system, Transp. Res. C 107 (2019) 287–300.
[35] P. Kollmeyer, Panasonic 18650PF Li-ion Battery Data, Mendeley Data, Version 1, 2018. http://dx.doi.org/10.17632/wykht8y7tg.1.
[36] M. Youmans, J.C. Spainhour, P. Qiu, Classification of antibacterial peptides using long short-term memory recurrent neural networks, IEEE/ACM Trans. Comput. Biol. Bioinform. 17 (4) (2019) 1134–1140.